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JOSÉ RIZAL UNIVERSITY

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EXAMINATION BOOK

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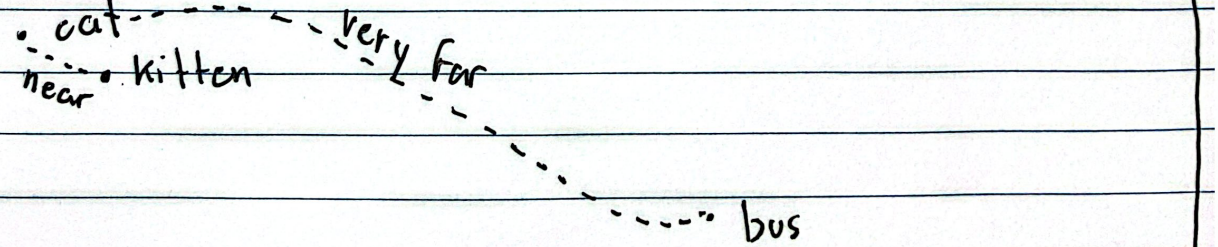
SUBJECT ITC C508 - 401I

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EXAMINATION BOOK

1.) Word Embeddings are a way to turn words into numbers to make computers understand how human language work. On a semantic space, these words are assigned coordinates called vectors. The relationship and similarities of these words are determined from their meanings and cosign similarities. If one word has similar meaning with another word they will be closer together, but if those two words have entirely different meanings they will be farther from each other.

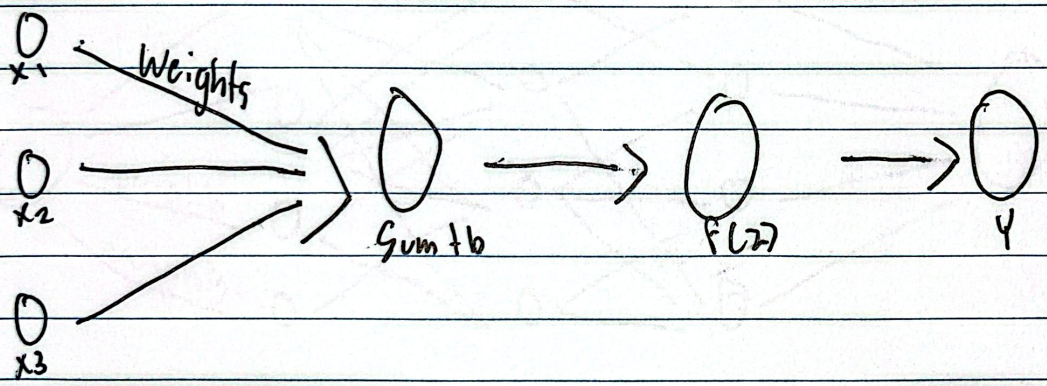
For example, on the lecture, the given was a "cat" and "kitten." Based on their meaning and how closely related they are, both of those words are closer together in the semantic space. when the given is "cat" and "bus," those two words are not related at all based on their meaning and category. This is how can an NLP model identify similarities and relationships between words and not just checking the spelling of the words.

Semantic space

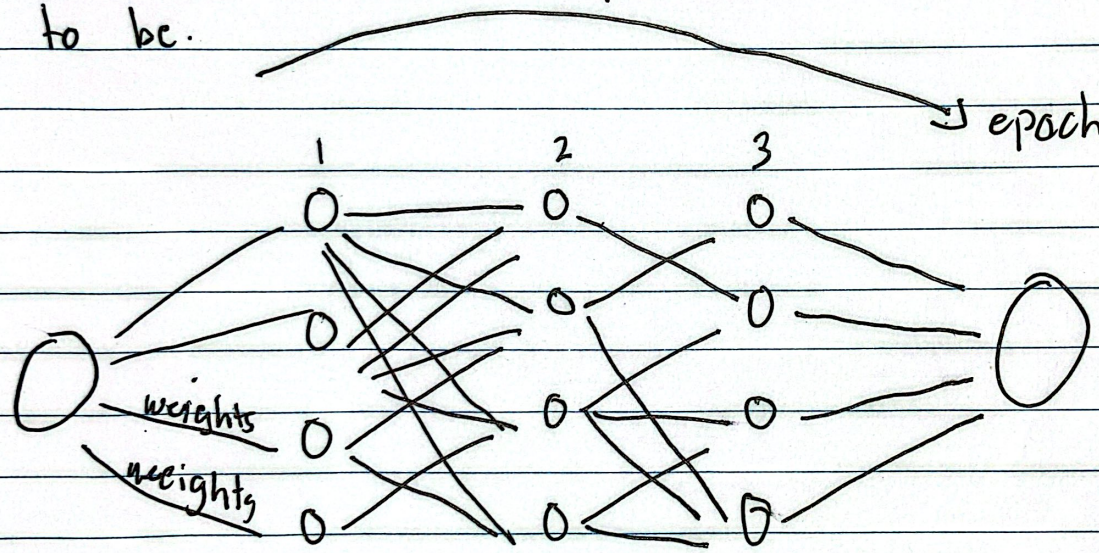


2.) a single neuron and linear regression equivalent to their inputs and weight given to each input. Through those inputs helps the neuron make predictions. through expanding network depth, and width each neuron can examine data at the same time.

When the each neuron can examine data at the same time they analyze complex patterns better because they are able to learn from previous layers. for example one neuron can learn a shape of a cat, then another can learn its color. Then that will result to all the layers being able to recognize patterns like its ears, eyes, and entire shape.



3.) Parameter's values are learned through training data such as weight values for the connection of input and hidden layers. through back propagation, it calculates its errors to learn better. While Hyperparameters are the settings we provide the model for better guidance such as, optimizer, number of epochs, number of hidden layers, how big is the data. For me, the actual learned intelligence is the parameter because of how it relearns each layer to come up with a better output. While the one that guides the model is the hyperparameter since we the user determines how indepth we want the model to be.



Parameter

- weights

Hyperparameter

- 3 hidden layers
- number of epoch : 15
- Optimizer : RELU